"Hierarchy of Needs" for Robots: Control Synthesis for Compositions of Hierarchical, Complex Objectives

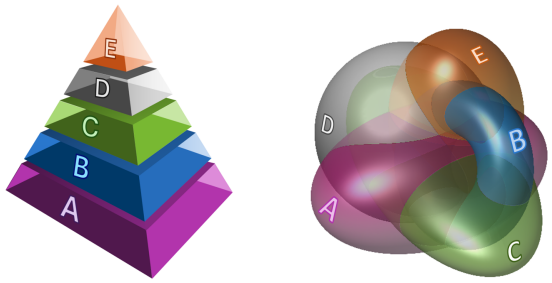
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***Abstract*— Drawing inspiration from Maslow's "hierarchy of needs", this paper develops a real-time control synthesis framework for robots to address hierarchical, complex objectives, recognizing that their behaviors are inherently driven by underlying needs. Each need is encoded by the zero-superlevel set of a control barrier function (CBF), which can be time-varying, and all the needs at the same level in a hierarchy are composed into a single one through Boolean compositions of the corresponding CBFs. The effectiveness of the proposed framework is demonstrated through a hypothetical interstellar exploration mission using laboratory robots, and new results on nonsmooth CBF and time-varying CBF are derived.**

I. Introduction

Robots are progressively expected to achieve ever more complex goals, [1], which poses a significant challenge in synthesis of behaviors that take into account diverse objectives involved. In addition, dynamic external factors, such as noise, perturbations, sensor inaccuracies, and actuator limitations, complicate the process by which robots achieve their objectives, necessitating that decisions are made autonomously and actions are executed in real time.

Pareto optimization, e.g., [2], balances multiple objectives effectively. However, in some missions, the diverse objectives may exhibit a hierarchical structure, where some should be prioritized, while some others may be considered supplementary, and in extreme cases, failure to satisfy certain objectives could lead to catastrophic consequences. Lexicographic optimization, e.g., [3], unlike Pareto optimization, can strictly prioritize multiple objectives, but this type of nested optimization framework may impose a computational burden on real-time robotic implementations. To address feasibility issues in control barrier function-based quadratic programs (CBF-QPs), a hierarchical relaxation framework is proposed in [4] to mainly deal with the conflicts between two constraints. However, generalizing this framework to handle multiple layers of constraints was not experimentally validated. Moreover, Boolean operations, AND ($\wedge$), OR ($\vee$), and NOT ($\neg$), on CBFs (see, e.g., [5], [6]) can compose multiple objectives effectively, but this method itself lacks an inherent mechanism to encode hierarchical architectures.



(a) (b)

Fig. 1. (a) A pyramid illustrating a hierarchy of needs for a robot; (b) The sets (which can be time-varying) in the state space of the robot, with each color corresponding to the need at a certain level in the hierarchy.

Remark I.1. *We cannot use the secondary Boolean operation $\neg(x \in \mathcal{S}_a) \rightarrow (x \in \mathcal{S}_b)$ to encode the proposition that the state x of a dynamical system should stay in the set $\mathcal{S}_a$, but if it cannot, then it should stay in the set $\mathcal{S}_b$ as a fallback. This is because $\neg(x \in \mathcal{S}_a) \rightarrow (x \in \mathcal{S}_b)$ is logically equivalent to $(x \in \mathcal{S}_a) \vee (x \in \mathcal{S}_b)$, which does not encode the priority of $\mathcal{S}_a$ over $\mathcal{S}_b$, i.e., $\mathcal{S}_a$ and $\mathcal{S}_b$ are equally important.*

Although synthesizing behaviors based on diverse, hierarchical objectives is common for humans, enabling robots to autonomously achieve this capability remains elusive [7]. Maslow's hierarchy of needs [8] suggests that human behavior is mainly driven by underlying needs. For example, when thirst occurs, which can be quantified by changes in the body's hydration levels, the behavior of seeking and consuming water is driven by the need for hydration. After the physiological needs are satisfied to some extent, behavior typically shifts toward pursuing higher-level needs such as safety, social connection, and self-actualization. Inspired by this idea, we propose that the behaviors of a robot should be similarly driven by (potentially hierarchical) needs, meaning that it does nothing unless there is a need.

To that end, we encode the need of a robot at each level in a hierarchy by a set in its state space, as illustrated in Fig. 1. The need/objective is said to be satisfied/achieved if the state is in the corresponding set. The state can encode a wide range of quantities of interest, which serve as representations of various needs. For example, in certain tasks, basic operational needs can be represented by quantities such as

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power and energy levels; perception needs by sensing range and field of view; safety needs by collision avoidance; and collaboration needs by algebraic connectivity.

To implement this idea into a robot-executable control policy, we rely on Boolean compositions of CBFs because the set encoding the need at each level in the hierarchy can be complex, necessitating Boolean compositions of multiple different sets. For example, at a certain level in a hierarchy, a robot needs to map its local environment and achieve a certain range of scene reconstruction, typically requiring the sensor range to be within a set. However, if the robot can communicate with others that have already mapped its local environment, it is no longer expected to achieve/satisfy the sensor range objective/need, and can thus pursue higher-level needs such as safe navigation using the constructed map. In such a case, the needs of sensing and communication can be composed by Boolean operation $\vee$.

Moreover, we interpret "hierarchy of needs" for robots from the perspective that when a robot satisfies the need at a certain level, it will be pursuing a higher-level need.

The main contributions of this paper are as follows:

(1) We show that in certain scenarios, using the method developed in [6], logically consistent Boolean operations on CBFs have different and even undesirable reachability implications. Then, we propose an extension of the method in [6] to address this issue;
(2) In addition to the set forward invariance property of the time-varying CBF, e.g., [9]–[11], we prove its attractivity properties in the sense that when the state is outside a time-varying set, it will asymptotically converge to the set, reach it within a finite time and remain on its boundary, or enter its interior within a finite time;
(3) We propose a control synthesis framework that can encode any number of hierarchical needs/objectives of a robot, enabling autonomous decision-making and real-time action execution; and
(4) The effectiveness of the proposed framework is demonstrated through a hardware-in-the-loop experiment using laboratory robots in a hypothetical interstellar exploration mission.

II. Composition of Complex Objectives

An arbitrary number of objectives/needs of a robot can be encapsulated into a single one through Boolean compositions of corresponding CBFs. Specifically, we consider throughout this paper the state of a robot to be in Euclidean space and governed by

$$\dot{x} = f(x) + g(x)u, \tag{1}$$

where $x \in \mathbb{R}^n$, $u \in \mathcal{U} \subset \mathbb{R}^m$, and the vector fields $f : \mathbb{R}^n \to \mathbb{R}^n$ and $g : \mathbb{R}^n \to \mathbb{R}^{n\times m}$ are locally Lipschitz.

According to [12], given a compact set $\mathcal{S} = \{x \in \mathbb{R}^n \,|\, h(x) \geq 0\}$ with its boundary defined as $\partial\mathcal{S} = \{x \in \mathbb{R}^n \,|\, h(x) = 0\}$, where $h \in C^1 : \mathbb{R}^n \to \mathbb{R}$, and $\nabla h(x) \neq 0$, $\forall x \in \partial\mathcal{S}$, h is a CBF with respect to (1) and $\mathcal{S}$ if there exists a locally Lipschitz extended class $\mathcal{K}_\infty$ function α as defined in [13] such that $\forall x \in \mathbb{R}^n$, $\sup_{u\in\mathcal{U}} \mathcal{C}(x, u; h) \geq 0$ with $\mathcal{C}(x, u; h) \coloneqq \nabla h(x)^\top f(x) + \nabla h(x)^\top g(x)u + \alpha(h(x))$, where the inequality $\mathcal{C}(x, u; h) \geq 0$ is called a control barrier certificate (CBC). Moreover, if h is a CBF, any locally Lipschitz continuous u that satisfies $\mathcal{C}(x, u; h) \geq 0$, $\forall x \in \mathcal{S}$, renders $\mathcal{S}$ forward invariant, e.g., [12], [13].

Suppose we have two CBFs, h_1 and h_2, based on [6],

$$\mathcal{S}_1 \cap \mathcal{S}_2 = \{x \in \mathbb{R}^n \mid \min\{h_1(x), h_2(x)\} \geq 0\},$$
$$\mathcal{S}_1 \cup \mathcal{S}_2 = \{x \in \mathbb{R}^n \mid \max\{h_1(x), h_2(x)\} \geq 0\}.$$

Let the nondifferentiable function $\mathfrak{h} : \mathbb{R}^n \to \mathbb{R}$, resulting from $\min$ or $\max$ operations on CBFs, be Lipschitz near x. Then, according to [6], the corresponding CBC is

$$\min\{\partial_c\mathfrak{h}(x)^\top f(x) + \partial_c\mathfrak{h}(x)^\top g(x)u + \alpha(\mathfrak{h}(x))\} \geq 0, \tag{2}$$

where $\min$ is taken over $\partial_c\mathfrak{h} : \mathbb{R}^n \to 2^{\mathbb{R}^n}$ [14], defined as

$$\partial_c\mathfrak{h}(x) \coloneqq \operatorname{co}\left\{\lim_{i\to\infty} \nabla\mathfrak{h}(\tilde{x}_i) \,\middle|\, \tilde{x}_i \to x, \tilde{x}_i \notin \mathcal{E} \cup \mathcal{D}_\mathfrak{h}\right\} \tag{3}$$

for any $\mathcal{E} \subset \mathbb{R}^n$ such that $\mu(\mathcal{E}) = 0$, where co denotes the convex hull of a set and μ denotes the Lebesgue measure of a set, and $\mathcal{D}_\mathfrak{h}$ with $\mu(\mathcal{D}_\mathfrak{h}) = 0$ is the set in which $\mathfrak{h}$ is nondifferentiable. However, (2) can be problematic, not from a logic vantage-point though, as introduced in Example II.1.

Example II.1. *Given two CBFs, h_a and h_b, the implication $\neg(h_a \geq 0) \to (h_b \geq 0)$ is logically equivalent to*

$$(h_a \geq 0) \vee ((-h_a \geq 0) \wedge (h_b \geq 0)), \tag{4}$$

which is also logically equivalent to

$$(h_a \geq 0) \vee (h_b \geq 0), \tag{5}$$

where $\mu(\mathcal{S}_a \cap \mathcal{S}_b) \neq 0$ with $\mathcal{S}_a = \{x \in \mathbb{R}^n \,|\, h_a \geq 0\}$ and $\mathcal{S}_b = \{x \in \mathbb{R}^n \,|\, h_b \geq 0\}$. The nonsmooth CBF and its 0-superlevel set corresponding to (4) *are $\mathfrak{h}_A = \max\{h_a, \min\{-h_a, h_b\}\}$ and $\mathcal{S}_A = \{x \in \mathbb{R}^n \,|\, \mathfrak{h}_A \geq 0\}$, respectively. The nonsmooth CBF and its 0-superlevel set corresponding to* (5) *are $\mathfrak{h}_B = \max\{h_a, h_b\}$ and $\mathcal{S}_B = \{x \in \mathbb{R}^n \,|\, \mathfrak{h}_B \geq 0\}$, respectively. Although $\mathcal{S}_A = \mathcal{S}_B$, $\exists\, \mathcal{S}_C \in 2^{\mathcal{S}_A}$ such that $\mu(\mathcal{S}_C) = 0$ and $\mathfrak{h}_A(x) = 0$, $\forall x \in \mathcal{S}_C$.*

The issue caused by this nonempty zero-Lebesgue-measure set $\mathcal{S}_C$ in Example II.1 for control synthesis through (2) is characterized in Proposition II.1.

Proposition II.1. *Given two sets $\mathcal{S}_1$ and $\mathcal{S}_2$ defined as the 0-superlevel sets of two CBFs h_1 and h_2 respectively, where $\mathcal{S}_1 \cap \mathcal{S}_2 \neq \varnothing$ and $\mu(\mathcal{S}_1 \cap \mathcal{S}_2) = 0$. We denote $\mathfrak{h} = \max\{h_1, h_2\}$, whose corresponding 0-superlevel set is $\mathcal{S}_1 \cup \mathcal{S}_2$. If $x(t_0) \in \mathcal{S}_1^\circ$, then under the constraint* (2)*, $\nexists\, t \in [t_0, \infty)$ such that $x(t) \in \mathcal{S}_2$.*

Proof. (By contradiction) If $x(t_0) \in \mathcal{S}_1^\circ$, assume that $\exists\, t \in (t_0, \infty)$ such that $x(t) \in \mathcal{S}_2$, i.e., $h_2(x(t)) \geq 0$. Thus, under this assumption, by continuity and forward invariance of $\mathcal{S}_1 \cup \mathcal{S}_2$, $\exists\, \tau \in (t_0, t]$ such that $\mathfrak{h}(x(\tau)) = 0$ and $\nabla h_1(x(\tau)) = -\nu\nabla h_2(x(\tau)) \neq 0$, where $\nu \in \mathbb{R}_{>0}$. Since α is an extended class $\mathcal{K}_\infty$ function, then $\alpha(\mathfrak{h}(x(\tau))) = 0$. Using Proposition 2.3.12 in [14], the CBC (2) leads

to $\dot{\mathfrak{h}}(x(t)) + \alpha(\mathfrak{h}(x(t))) \geq 0$ almost everywhere. Since $\mathfrak{h}(x(t_0)) > 0$, then $\mathfrak{h}(x(t)) > 0$, $\forall t \in [t_0, \infty)$, which means that the assumption does not hold, and therefore Proposition II.1 is proved. □

Hence, when a set is constructed through Boolean compositions of CBFs involving at least one $\vee$, the situation described above may occur, using the method proposed in [6]. When it happens, the reachability to certain sets is prevented, and thus the robot may never fulfill higher-level needs, i.e., never reach the corresponding sets, even if the current level of need is satisfied. As such, the interpretation of "hierarchy of needs" for robots, discussed in Section I, cannot be achieved.

Therefore, we propose a new CBC, extended from (2),

$$\max\{\partial_c \mathfrak{h}(x)^\top f(x) + \partial_c \mathfrak{h}(x)^\top g(x)u + \alpha(\mathfrak{h}(x))\} \geq 0, \quad (6)$$

where $\max$ is taken over $\partial_c \mathfrak{h}$ defined by (3). The constraint (6) results in a larger set of available controllers, which may address the issue characterized in Proposition II.1 in the sense that when the state reaches such $\mathcal{S}_1 \cap \mathcal{S}_2$, the available controllers given by (6) allows the increase of either h_1 or h_2. For space efficiency, we omit the proof of the proposition on (6), which follows from a similar analysis procedure to the proof of Proposition II.1.

An alternative route, pursued in this paper, is to leverage smooth approximations, such as polynomial smoothing, e.g., [15], or log-sum-exp smoothing, e.g., [10], [16], to address the issue described in Example II.1 and Proposition II.1.

Note that a need can change over time. For example, the connectivity maintenance distance in multi-robot collaboration can vary in response to changing signal strength to ensure stable communication. Thus, we consider the time-varying set with its boundary and interior defined as

$$\begin{aligned} \mathcal{S}(t) &= \{x \in \mathbb{R}^n \,|\, h(x,t) \geq 0\}, \\ \partial\mathcal{S}(t) &= \{x \in \mathbb{R}^n \,|\, h(x,t) = 0\}, \\ \mathcal{S}^\circ(t) &= \{x \in \mathbb{R}^n \,|\, h(x,t) > 0\}, \end{aligned} \quad (7)$$

where $h \in C^1 : \mathbb{R}^n \times \mathbb{R}_{\geq 0} \to \mathbb{R}$, and $\frac{\partial h}{\partial x}(x,t) \neq 0$, $\forall x \in \partial\mathcal{S}$, $\forall t \in [t_0, \infty)$.

Definition II.1. (Definition 4 in [11] following [9], [10]) *$h(x,t)$ is a time-varying CBF (TV-CBF) with respect to* (1) *and* (7)*, if there exists a locally Lipschitz extended class $\mathcal{K}_\infty$ function α such that*

$$\sup_{u \in \mathcal{U}} \mathcal{C}(x,u,t;h) \geq 0,$$

$\forall t \in [t_0, \infty)$, where

$$\begin{aligned} \mathcal{C}(x,u,t;h) \coloneqq & \frac{\partial h}{\partial t}(x,t) + \frac{\partial h}{\partial x}(x,t)^\top f(x) \\ & + \frac{\partial h}{\partial x}(x,t)^\top g(x)u + \alpha(h(x,t)). \end{aligned} \quad (8)$$

According to [10], [17], a smooth under-approximation of $\min\{h_1(x,t), h_2(x,t)\}$, denoted as

$$h_{1\wedge 2}(x,t) \coloneqq h_1(x,t) \wedge h_2(x,t), \quad (9)$$

can be given by

$$h_{1\wedge 2}(x,t) = \frac{-1}{\kappa_{1\wedge 2}} \ln(e^{-\kappa_{1\wedge 2} h_1(x,t)} + e^{-\kappa_{1\wedge 2} h_2(x,t)}), \quad (10)$$

where $\kappa_{1\wedge 2} \in \mathbb{R}_{>0}$. In addition, a smooth over-approximation of $\max\{h_1(x,t), h_2(x,t)\}$, denoted as

$$h_{1\vee 2}(x,t) \coloneqq h_1(x,t) \vee h_2(x,t), \quad (11)$$

can be given by

$$h_{1\vee 2}(x,t) = \frac{1}{\kappa_{1\vee 2}} \ln(e^{\kappa_{1\vee 2} h_1(x,t)} + e^{\kappa_{1\vee 2} h_2(x,t)}), \quad (12)$$

where $\kappa_{1\vee 2} \in \mathbb{R}_{>0}$.

Note that although Lemma 1 in [11] implies that $\mathcal{S}(t)$ defined in (7) is rendered forward invariant by any locally Lipschitz continuous controller u that satisfies the time-varying CBC (TV-CBC), $\mathcal{C}(x,u,t;h) \geq 0$, based on the definition in (8), the initial state $x(t_0)$ is not necessarily in $\mathcal{S}(t_0)$. Moreover, other factors, such as noise or latency from actuators or sensors, perturbations from the environment, and the evolution of $\mathcal{S}(t)$ itself (if sometimes beyond the maximum capability of actuators) may cause the state x to be outside certain $\mathcal{S}(t)$ even if $x(t_0) \in \mathcal{S}(t_0)$.

As a key contribution of this paper, we investigate theoretical guarantees on the behaviors of the dynamical system (1) subject to the TV-CBC, which also supports the development of the framework in Section III.

To this end, first, building on Definition 4.3 in [18], we introduce Definition II.2.

Definition II.2. *A continuous function $\xi : \mathbb{R} \times \mathbb{R}_{\geq 0} \to \mathbb{R}$ is said to be an extended class $\mathcal{K}_\infty\mathcal{L}$ function if:* (a) *For each fixed s, $\xi(r,s)$ is strictly increasing with respect to r and $\xi(0,s) = 0$; and* (b) *For each fixed r, if $r \in \mathbb{R}_{>0}$, then $\xi(r,s)$ is strictly decreasing with respect to s and $\lim_{s\to\infty} \xi(r,s) = 0$; if $r = 0$, then $\xi(r,s) \equiv 0$, $\forall s \in \mathbb{R}_{\geq 0}$; if $r \in \mathbb{R}_{<0}$, then $\xi(r,s)$ is strictly increasing with respect to s and $\lim_{s\to\infty} \xi(r,s) = 0$.*

Then, we extend Lemma 4.4 in [18] to obtain Lemma II.1.

Lemma II.1. *The differential equation $\dot{\rho} + \alpha(\rho) = 0$ with $\rho(t_0) = \rho_0$, where α is a locally Lipschitz extended class $\mathcal{K}_\infty$ function, has a unique solution $\rho(t) = \xi(\rho_0, t - t_0)$, $\forall \rho_0 \in \mathbb{R}$, $\forall t \in [t_0, t_0 + \delta t_{max}(\rho_0))$, where ξ is an extended class $\mathcal{K}_\infty\mathcal{L}$ function per* Definition II.2*.*

Proof. See Appendix A (inspired by [18]). □

Using Lemma II.1, we establish Theorem II.1.

Theorem II.1. (Attractivity of TV-CBF) *If $h(x,t)$ is a TV-CBF per* Definition II.1*, then any locally Lipschitz continuous controller u satisfying the TV-CBC, $\mathcal{C}(x,u,t;h) \geq 0$, $\forall t \in [t_0, \infty)$, renders $\mathcal{S}(t)$ attractive in the sense that*

$$x(t_0) \in \mathcal{S}(t_0) \Rightarrow x(t) \in \mathcal{S}(t), \forall t \in [t_0, \infty) \quad (13)$$

$$x(t_0) \notin \mathcal{S}(t_0) \Rightarrow \lim_{t\to\infty} h(x(t),t) = 0 \quad (14)$$

or

$$\exists\, \tau \in (t_0, \infty), \text{ s.t. } x(t) \in \mathcal{S}(t), \quad \forall t \in [\tau, \infty), \quad (15)$$

where (15) *incorporates two situations: (a)* $\exists \tau \in (t_0, \infty)$ *such that* $x(t) \in \partial\mathcal{S}(t)$, $\forall t \in [\tau, \infty)$*; or (b)* $\exists \tau \in (t_0, \infty)$ *such that* $x(\tau) \in \partial\mathcal{S}(\tau)$ *and* $x(t) \in \mathcal{S}^\circ(t)$, $\forall t \in (\tau, \infty)$.

Proof. Since $h(x,t)$ is a TV-CBF, then its corresponding TV-CBC, $\mathcal{C}(x,u,t;h) \geq 0$, can be expressed as $\dot{h}(x,t,u) + \alpha(h(x,t)) \geq 0$. Consider a scalar differential equation $\dot{\rho} + \alpha(\rho) = 0$ with $\rho(t_0) = h(x(t_0), t_0)$. According to Lemma II.1, assuming forward completeness, $\dot{\rho} + \alpha(\rho) = 0$ has a unique solution $\rho(t) = \xi(h(x(t_0), t_0), t - t_0)$, $\forall t \in [t_0, \infty)$, where ξ is an extended class $\mathcal{K}_\infty\mathcal{L}$ function per Definition II.2. Based on Lemma 3.4 in [18], $h(x(t), t) \geq \xi(h(x(t_0), t_0), t - t_0)$, $\forall t \in [t_0, \infty)$.

When $h(x(t_0), t_0) > 0$, i.e., $x(t_0) \in \mathcal{S}^\circ(t_0)$, per Definition II.2, $\xi(h(x(t_0), t_0), t - t_0)$ is strictly decreasing with respect to t and $\lim_{t\to\infty} \xi(h(x(t_0), t_0), t - t_0) = 0$, so $h(x(t), t) > 0$, $\forall t \in [t_0, \infty)$, which proves (13).

When $h(x(t_0), t_0) < 0$, i.e., $x(t_0) \notin \mathcal{S}(t_0)$, per Definition II.2, $\xi(h(x(t_0), t_0), t - t_0)$ is strictly increasing with respect to t and $\lim_{t\to\infty} \xi(h(x(t_0), t_0), t - t_0) = 0$. Since $\xi(h(x(t_0), t_0), t - t_0)$ is a lower bound of $h(x(t), t)$ and asymptotically converges to 0, the evolution of $h(x(t), t)$ with respect to t is such that either (14) or (15). □

Some results from Theorem II.1 recover certain existing work, where (13) is the forward invariance property proved in [11] based on [19], and (14) corresponds to the robustness or asymptotic stability property proved in [20] for the time-invariant CBF.

Moreover, thanks to the attractivity of TV-CBF per Theorem II.1, even if the state is outside the set due to factors such as initial conditions, noise, and networ latency, it will be attracted to the set in one of the three manners described in Theorem II.1.

III. Composition of Hierarchical Needs

Using the techniques discussed in Section II, we can compose an arbitrary number of TV-CBFs through basic Boolean operations to obtain a single TV-CBF, whose 0-superlevel set in the state space encodes the composition of complex objectives/needs.

It should be noted that, as discussed in Section I, the diverse objectives/needs can come from various contexts, such as the need of operational power, torque level, battery temperature range, energy storage and consumption, and information flow rate. Consequently, the corresponding TV-CBFs may have different orders of magnitude. To avoid potential numerical issues e.g., [16], we can apply a scaling function $\sigma : \mathbb{R} \to (a_\sigma, b_\sigma)$ on a TV-CBF, where $a_\sigma \in \mathbb{R}_{<0}$, $b_\sigma \in \mathbb{R}_{>0}$, $\sigma(0) = 0$, and σ is strictly increasing with respect to its argument. In this paper, we choose to scale a TV-CBF, if necessary, by a sigmoid-like function

$$\tilde{h}(x,t) \coloneqq \sigma(h(x,t)) = \frac{2}{1 + e^{-wh(x,t)}} - 1, \qquad (16)$$

where $w \in \mathbb{R}_{>0}$. The scaled TV-CBF $\tilde{h}(x,t)$ results in the same 0-superlevel set as $h(x,t)$ and also has the attractivity properties discussed in Theorem II.1. To make efficient use of space, we omit the proof of this proposition on (16), which can be obtained by following a similar procedure to that of the proof of Theorem II.1.

Nonetheless, only manipulating the sets by Boolean operations cannot encode the composition of hierarchical objectives/needs, as discussed in Section I.

Let the set of indices of all the TV-CBFs associated with a robot be $\mathcal{P} = \{1, 2, \ldots, P\}$, where $P \in \mathbb{Z}^+$. Each TV-CBF is denoted as $h_p(x,t)$, $\forall p \in \mathcal{P}$. In a hierarchy of needs for the robot, there are $L \in \mathbb{Z}^+$ levels of needs, each of which is encoded by a single TV-CBF, $H_\ell(x,t)$, $\forall \ell \in \mathcal{L} = \{1, 2, \ldots, L\}$, which is a smooth approximation of an arbitrary number of Boolean compositions of $h_q(x,t)$ (or $\tilde{h}_q(x,t)$ if scaled by (16)), $\forall q \in \mathcal{Q}_\ell \subseteq \mathcal{P}$, where $\mathcal{Q}_\ell$ represents the set of indices of the TV-CBFs at Level ℓ. Typically, $\cap_{\ell\in\mathcal{L}} \mathcal{Q}_\ell = \varnothing$ and $\cup_{\ell\in\mathcal{L}} \mathcal{Q}_\ell = \mathcal{P}$. Additionally, we assume that $H_1(x,t)$ encodes the lowest level of need and $H_L(x,t)$ encodes the highest level of need.

As such, based on the "hierarchy of needs" for robots interpreted in Section I, a robot is programmed to do nothing, subject to its hierarchical needs, encoded by the following control synthesis framework.

$$\begin{aligned} \arg\min_{u,\delta_{\ell+1}} \ & \|u\|^2 + \sum_{\ell\in\tilde{\mathcal{L}}} \kappa_{\ell+1}\delta_{\ell+1}^2 \\ \text{s.t.} \ & \mathcal{C}(x,u,t;\tilde{H}_1) \geq 0 \\ & \prod_{l=1}^{\ell} \Phi(\tilde{H}_l) \cdot \mathcal{C}(x,u,t;\tilde{H}_{\ell+1}) \geq \delta_{\ell+1}, \ \forall \ell \in \tilde{\mathcal{L}}, \end{aligned} \qquad (17)$$

where $\tilde{\mathcal{L}} \coloneqq \mathcal{L} \setminus \{L\}$, and $\delta_{\ell+1} \in \mathbb{R}$ serves as a relaxation variable penalized by $\kappa_{\ell+1} \in \mathbb{R}_{>0}$ to prevent potential conflict among different constraints.

Remark III.1. *At a moment in time, whether the state* x *is in the* 0*-superlevel set of a CBF has no causal relation with whether there exists a solution to the corresponding CBF-based optimization problem (without relaxation variables).*

In addition, the indicator function $\Phi(\tilde{H}_l)$ is defined such that the need at a certain level is pursued when all of its lower-level needs are satisfied, and its choice, similar to the scaling function (16), is not unique. In this paper, we let $\Phi(\tilde{H}_l) = 1$ if $\tilde{H}_l > b_\phi$; $\Phi(\tilde{H}_l) = 3\left(\frac{\tilde{H}_l - a_\phi}{b_\phi - a_\phi}\right)^2 - 2\left(\frac{\tilde{H}_l - a_\phi}{b_\phi - a_\phi}\right)^3$ if $\tilde{H}_l \in [a_\phi, b_\phi]$; and $\Phi(\tilde{H}_l) = 0$ if $\tilde{H}_l < a_\phi$, in which $a_\phi \to 0^-$ serves as a margin to handle the case of (14), and $b_\phi > a_\phi$.

Note that when a robot has not satisfied a lower-level need, it may still satisfy a higher-level need by chance, even though it is not actively pursuing this higher-level need due to the term $\prod_{l=1}^{\ell} \Phi(\tilde{H}_l)$ in (17). Moreover, since (17) is a quadratic program, it can be executed by robots in real time.

IV. Example Scenario and Experiment

Consider a hypothetical scenario where interstellar exploration robots are deployed on an extraterrestrial planet whose terrain shifts dynamically, with vast regions covered in plasma fields hazardous to the robots. However, some

safe zones remain accessible to robots, although the configurations of these zones may vary as the terrain changes. Moreover, certain highlands in these zones, due to the higher density of extraterrestrial flora and fauna, are particularly suitable for ecological monitoring and research. Furthermore, maintaining connectivity is essential for robots to efficiently share information and collaboratively reach these highlands. In addition, roaming electromagnetic storms pose a threat by potentially impairing the functionality of robots.

We apply the proposed framework (17) to this hypothetical interstellar exploration mission using differential-drive wheeled robots in a rectangular area $\mathcal{A} \subset \mathbb{R}^2$ with x-axis ranging from -1.6 to 1.6 meters and y-axis ranging from -1 to 1 meters as shown in Fig. 2. The state of a robot is defined as the position $x_i \in \mathbb{R}^2$ of a point near, but not on, its axle, and the single-integrator dynamics $\dot{x}_i = u_i$ is assumed $\forall i \in \mathcal{I} = \{1, 2, \ldots, I\}$, which is transformed into executable controllers by the near-identity diffeomorphism [21]. Specifically, we consider $I = 3$ homogeneous robots in the sense that they have the same hierarchy of needs with $L = 5$ levels. For each robot, their CBFs, $h_{i,p}(x_i, t)$, $\forall p \in \mathcal{P} = \{1, 2, \ldots, 22\}$, are defined as follows.

The plasma fields can cause catastrophic damage to robots, so the need at the bottom of the hierarchy of needs for each robot is to stay in the safe zones, while avoiding collisions with other robots as required in physical robotic experiments. Based on (9), (10), (11), (12), let the smooth approximation of the TV-CBF at Level 1 in the hierarchy be

$$\begin{aligned} H_1(x_i, t) = &((h_{i,1} \wedge h_{i,2} \wedge h_{i,3} \wedge h_{i,4}) \vee \\ &(h_{i,5} \wedge h_{i,6} \wedge h_{i,7} \wedge h_{i,8}) \vee \\ &(h_{i,9} \wedge h_{i,10} \wedge h_{i,11} \wedge h_{i,12} \wedge h_{i,13} \wedge h_{i,14})) \wedge \\ &h_{i,15} \wedge h_{i,16}, \end{aligned} \tag{18}$$

in which $h_{i,p}(x_i, t) = a_p^\top x_i + b_p(t)$ with $a_p \in \mathbb{R}^2$ and $b_p(t) \in \mathbb{R}$, $\forall p \in \{1, 2, \ldots, 14\}$. The TV-CBF (18) encodes that each robot should stay in Zone Z1 (the purple trapezoid), Zone Z2 (the purple rectangle), or Zone Z3 (the purple hexagon), as illustrated in Fig. 2(a), while avoiding collisions with other robots, where the CBFs for collision avoidance are given by

$$h_{i,15}(x_i, t) = \|x_i - x_j(t)\|^2 - d_{\text{CA}}^2, \tag{19}$$

$$h_{i,16}(x_i, t) = \|x_i - x_k(t)\|^2 - d_{\text{CA}}^2, \tag{20}$$

in which $i \neq j \neq k \in \mathcal{I}$ and $d_{\text{CA}} \in \mathbb{R}_{>0}$. The CBF for collision avoidance is formulated in the time-varying manner to enable both Boolean compositions and decentralized controller synthesis. This cannot be achieved by the decentralized quadratic program for collision avoidance in [22], based on the formulation in [23], because the constraint in the decentralized quadratic program in [22] is not a CBC.

The potential impairment of robots caused by electromagnetic storms is less severe than the damage caused by plasma fields, so the TV-CBF at Level 2 is given by

$$H_2(x_i, t) = h_{i,17} \wedge h_{i,18}, \tag{21}$$

in which $h_{i,17}(x_i, t) = \|x_i - x_{\text{ES},1}(t)\|^2 - d_{\text{ES}}^2$ and $h_{i,18}(x_i, t) = \|x_i - x_{\text{ES},2}(t)\|^2 - d_{\text{ES}}^2$ with $d_{\text{ES}} \in \mathbb{R}_{>0}$, where $x_{\text{ES},1}(t)$ and $x_{\text{ES},2}(t)$ are the center positions of two electromagnetic storms. $H_2(x_i, t)$ is hierarchically above $H_1(x_i, t)$, meaning that if the robot can stay in the safe zones and avoid collisions, it should try to avoid the electromagnetic storms, represented by the two blue circular regions in Fig. 2.

When the robot satisfies its safety needs, it will consider connecting with other robots, similar to social needs. Thus, the TV-CBF at Level 3 is defined as

$$H_3(x_i, t) = h_{i,19} \wedge h_{i,20}, \tag{22}$$

where $h_{i,19}(x_i, t) = d_{\text{C}}^2 - \|x_i - x_j(t)\|^2$ and $h_{i,20}(x_i, t) = d_{\text{C}}^2 - \|x_i - x_k(t)\|^2$ with $d_{\text{C}} \in \mathbb{R}_{>0}$, in which $i \neq j \neq k \in \mathcal{I}$, encoding connectivity maintenance with other robots, formulated in a time-varying manner due to the same reason as the TV-CBFs for collision avoidance.

Once connected with other robots, the robot proceeds toward the highland with a higher density of extraterrestrial flora and fauna, encoded by the CBF at Level 4,

$$H_4(x_i) = h_{i,21}(x_i), \tag{23}$$

where $h_{i,21}(x_i) = r_{\text{H}}^2 - \|x_i - o_{\text{H}}\|^2$ with $r_{\text{H}} \in \mathbb{R}_{>0}$ and $o_{\text{H}} \in \mathbb{R}^2$. The highland is enclosed by the gray circle, as shown in Fig. 2.

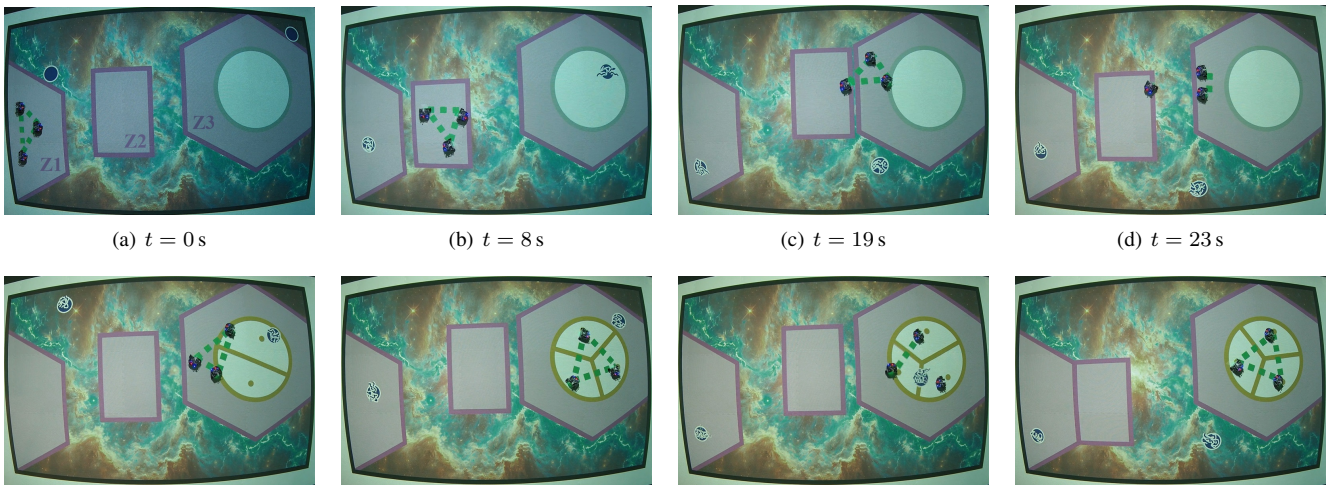
Once the robot reaches the highlands, it will strive to optimally cover the area for ecological monitoring of extraterrestrial flora and fauna. Hence, the CBF at Level 5 can be given by

$$H_5(x_i) = h_{i,22}(x_i) = -\frac{1}{2}\|x_i - c_i\|^2, \tag{24}$$

e.g., [24], where $c_i = (\int_{\mathcal{V}_i} z\psi(z)\,dz)/(\int_{\mathcal{V}_i} \psi(z)\,dz)$ is the center of mass of the Voronoi cell of x_i, which is given by $\mathcal{V}_i(x) = \{z \in \mathcal{A} \mid \|x_i - z\| \leq \|x_j - z\|, \forall j \neq i \in \mathcal{I}\}$ with $x = [x_1^\top, \ldots, x_I^\top]^\top$ and $c = [c_1^\top, \ldots, c_I^\top]^\top$. Additionally, $\psi(z) = e^{-10\Psi(\|x_i - o_{\text{H}}\|^2 - r_{\text{H}}^2)}$, in which $\Psi(y) = y(\tan^{-1}(10y)/\pi + 0.5)$, $\forall y \in \mathbb{R}$, e.g., [25].

We scale the resulting single CBFs at each level (18), (21), (22), (23), and (24) by (16) and then substitute them into (17) based on (8), which is then implemented in a hardware-in-the-loop experiment, whose snapshots are shown in Fig. 2.

The three robots are initialized in Zone Z1, as shown in Fig. 2(a). Since the needs at Levels 1 to 3 are satisfied, the robots are pursuing the needs at Level 4, i.e., trying to reach the highland, through Zone Z2 that is time-varying, as shown in Fig. 2(b). However, as seen in Fig. 2(c), a robot did not get a chance to enter Zone Z3 this time, so the robot stays in Zone Z2. Meanwhile, as seen in Fig. 2(d), each of the other two robots has lost connectivity with the robot in Zone Z2, meaning that the need at Level 3 is not satisfied for each robot. As a result, they are not pursuing the needs at Level 4, i.e., reaching the highland. Although the two robots in Zone Z3 are trying to connect with the robot in Zone Z2 that is also trying to connect with the robots in Zone Z3, they cannot achieve the needs at Level 3 due to their needs at Level 1, but they have to stay in safe zones at least, while striving to satisfy the needs at Level 2. When the robots enter Zone Z3 and remain connected, as shown in Fig. 2(e), they proceed



(a) $t = 0\,\mathrm{s}$ (b) $t = 8\,\mathrm{s}$ (c) $t = 19\,\mathrm{s}$ (d) $t = 23\,\mathrm{s}$

(e) $t = 44\,\mathrm{s}$ (f) $t = 58\,\mathrm{s}$ (g) $t = 65\,\mathrm{s}$ (h) $t = 73\,\mathrm{s}$

Fig. 2. Snapshots of the hardware-in-the-loop experiment in the hypothetical interstellar exploration mission using three differential-drive wheeled robots, where the extraterrestrial environment is represented through projected scenes. The purple zones Z1, Z2, and Z3 represent the safe zones in which the robots must remain to prevent entry into the plasma fields. The two blue circular regions with lightning represent the electromagnetic storms. When two robots are connected, a green dashed line appears between them. The highland, with a higher density of extraterrestrial flora and fauna, is enclosed by the gray circle. This circle turns orange when at least two robots reach the highland. The orange lines in the highland represent the boundaries of the Voronoi cells of the robots and the orange dots represent their corresponding centers of mass, i.e., optimal coverage locations.

to reach the highland and then pursue optimal coverage of the highland for ecological monitoring of extraterrestrial flora and fauna, as shown in Fig. 2(f). To avoid the electromagnetic storm, robots can lose connectivity with each other, as seen in Fig. 2(g), since the need at Level 2 has priority over the need at Level 3. After the electromagnetic storm moves away, the robots resume optimally covering the highland for ecological monitoring, as seen in Fig. 2(h).

V. Conclusion

In this paper, we identify a situation where the nonsmooth approach proposed in [6] causes undesirable reachability implications. Then, we suggest an extension of the method in [6] to address this issue. Moreover, we prove the attractivity properties of TV-CBF. Furthermore, we propose a real-time minimum-energy control synthesis framework encoding the "hierarchy of needs" for robots, under which a robot can make decisions and execute actions in real time driven by its hierarchical needs. The effectiveness of this proposed framework is demonstrated experimentally.

VI. Appendix A
Proof of Lemma II.1

Proof. Since α is locally Lipschitz, per Cauchy-Lipschitz theorem, $\dot{\rho} = -\alpha(\rho)$ with $\rho(t_0) = \rho_0$ has a unique solution, $\forall \rho_0 \in \mathbb{R}$, $\forall t \in [t_0, t_0 + \delta t_{max}(\rho_0)]$, and $\delta t_{max} = \infty$ if the differential equation is forward complete.

Integrating $\dot{\rho} = -\alpha(\rho)$ results in $\int_{\rho_0}^{\rho} \frac{d\rho}{\alpha(\rho)} = -\int_{t_0}^{t} dt$. When $\rho_0 \in \mathbb{R}_{>0}$, since α is an extended class $\mathcal{K}_\infty$ function, then $\dot{\rho}(t) \in \mathbb{R}_{<0}$, $\rho(t) \in \mathbb{R}_{>0}$, $\forall t \in [t_0, t_0 + \delta t_{max}(\rho_0)]$. Let $\epsilon_+ \in \mathbb{R}_{>0}$ be an intermediate variable and define $\varphi_+(\rho) = -\int_{\epsilon_+}^{\rho} \frac{d\rho}{\alpha(\rho)}$. Then we have $\varphi_+(\rho(t)) - \varphi_+(\rho_0) = \int_{\rho}^{\rho_0} \frac{d\rho}{\alpha(\rho)} = t - t_0$. Thus, $\varphi_+ : (0, \rho_0] \to [\varphi_+(\rho_0), \infty)$ is strictly decreasing with respect to ρ and hence $\varphi_+^{-1} : [\varphi_+(\rho_0), \infty) \to (0, \rho_0]$ exists. Therefore, $\rho(t) = \varphi_+^{-1}(\varphi_+(\rho_0) + t - t_0)$, $\forall \rho_0 \in \mathbb{R}_{>0}$, $\forall t \in [t_0, t_0 + \delta t_{max}(\rho_0)]$.

Likewise, when $\rho_0 \in \mathbb{R}_{<0}$, we have $\dot{\rho}(t) \in \mathbb{R}_{>0}$, $\rho(t) \in \mathbb{R}_{<0}$, $\forall t \in [t_0, t_0 + \delta t_{max}(\rho_0)]$. We define $\varphi_-(\rho) = -\int_{\epsilon_-}^{\rho} \frac{d\rho}{\alpha(\rho)}$ with $\epsilon_- \in \mathbb{R}_{<0}$ as an intermediate variable. Thus, $\varphi_- : [\rho_0, 0) \to [\varphi_-(\rho_0), \infty)$ is strictly increasing with respect to ρ and hence $\varphi_-^{-1} : [\varphi_-(\rho_0), \infty) \to [\rho_0, 0)$ exists. Therefore, $\rho(t) = \varphi_-^{-1}(\varphi_-(\rho_0) + t - t_0)$, $\forall \rho_0 \in \mathbb{R}_{<0}$, $\forall t \in [t_0, t_0 + \delta t_{max}(\rho_0)]$. Besides, when $\rho_0 = 0$, $\rho(t) \equiv 0$. Then we define

$$\xi(r, s) = \begin{cases} \varphi_+^{-1}(\varphi_+(r) + s), & r \in \mathbb{R}_{>0} \\ 0, & r = 0 \\ \varphi_-^{-1}(\varphi_-(r) + s), & r \in \mathbb{R}_{<0}. \end{cases}$$

Using the inverse function rule and Leibniz integral rule, according to the properties of φ_+^{-1} and φ_-^{-1}, we have

$$\frac{\partial \xi}{\partial r} = \begin{cases} \frac{\varphi_+'(r)}{\varphi_+'(\xi(r,s))} = \frac{\alpha(\xi(r,s))}{\alpha(r)} \in \mathbb{R}_{>0}, & r \in \mathbb{R}_{>0} \\ 0, & r = 0 \\ \frac{\varphi_-'(r)}{\varphi_-'(\xi(r,s))} = \frac{\alpha(\xi(r,s))}{\alpha(r)} \in \mathbb{R}_{>0}, & r \in \mathbb{R}_{<0}, \end{cases}$$

$$\frac{\partial \xi}{\partial s} = \begin{cases} \frac{1}{\varphi_+'(\xi(r,s))} = -\alpha(\xi(r,s)) \in \mathbb{R}_{<0}, & r \in \mathbb{R}_{>0} \\ 0, & r = 0 \\ \frac{1}{\varphi_-'(\xi(r,s))} = -\alpha(\xi(r,s)) \in \mathbb{R}_{>0}, & r \in \mathbb{R}_{<0}, \end{cases}$$

where $\varphi'(\cdot)$ denotes the first-order derivative of $\varphi(\cdot)$ with respect to its argument, so $\rho(t) = \xi(\rho_0, t - t_0)$, $\forall \rho_0 \in \mathbb{R}$, and ξ is an extended class $\mathcal{K}_\infty\mathcal{L}$ function per Definition II.2. □

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